

Reproduction of tysam-code/hlb-CIFAR10: hlb-CIFAR10 speed training

K-Veritas Team

Submitted by K-Veritas Team

kv:2610.00005v1 · Published 2026-10-07

Independent reproduction · Benchmark or leaderboard entry · Computer vision

Abstract

Ran main.py unmodified at the pinned commit: the script's default 25 independent CIFAR-10 training runs of the hlb-CIFAR10 network (about 12 epochs each, label smoothing, EMA), reporting per-epoch losses and accuracies and the mean and variance of the final EMA validation accuracy across runs. CIFAR-10 downloaded by torchvision. Single NVIDIA RTX 5060 Ti 16 GB, PyTorch 2.11 cu128, Python 3.12.

About this record

Title, authors and abstract were provided by the authors and reviewed for completeness, not for scientific soundness. Every number in this record is copied from the signed data of the sealed reports [2], which remain authoritative.

This cover page is not signed. Check it against the SHA-256 listed at <https://kveritas.org/records/2610.00005v1>, and verify each report with its code bundle: `kveritas verify <report.pdf> --bundle <bundle.zip>`.

1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	hlb-CIFAR10 (25 runs)	2026-10-07	1	1	Signed by the K-Veritas server

2 Results

r1: hlb-CIFAR10 (25 runs)

Run	Metric	Final value	Step
1	val_acc	0.9396	r25_e12
1	final_ema_val_acc	0.9407	25
1	total_time_seconds	21.9814	25

Run	Inline claim	Value	Phase
1	mean_final_ema_val_acc	0.940356	
1	var_final_ema_val_acc	2.03171e-06	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	705.9 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	218750b2e435899face808ae995d08e39e39733b5b6a697d011eab517959258f	6.5 MB
r1.kvbundle.zip	25e4fdd3df47d58c3328c744f9006bf3e78dbbca1c6f63220909851d2da934fe	16 KB

5 Links and licence

Reproduction of: <https://github.com/tysam-code/hlb-CIFAR10/tree/993965202fc4dc05b699a39db33c00f1c7f647fd>

Record: CC-BY 4.0. Code: Apache-2.0, as declared by the authors. Tags: cifar-10, speedrun, image-classification.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (hlb-CIFAR10 (25 runs)), kv:2610.00005v1/r1. Data hash d56329f8bb8a9415d4351f62343050154749361a15e83760f1f4c87f561086f6.

<https://kveritas.org/records/2610.00005v1/r1.pdf>